MNIST and CIFAR-10: native-workflow timing comparison

18 October 2026

Non-IID, seed 42, 100 initial clients, 40% FR3. Our v15.4 data: 100 completed rounds per dataset. STD-DAGMM: 100 completed rounds per dataset, five fresh fitting epochs each, no removals. These totals are measured, not projected. Components absent from historical logs are not estimated. Network communication was not measured. Different native trajectories, hardware load, local attacker evaluation and timer boundaries preclude a causal speedup comparison. The CIFAR detector encountered near-zero available RAM, so paging may affect its timings; paging was not measured.

Table 1: Measured STD-DAGMM component times (seconds per round, mean ± sample SD, 100 rounds per dataset).

Component	MNIST	CIFAR-10
Client training	33.986 ± 1.539	76.208 ± 4.164
Simulated FR response	0.062 ± 0.006	0.255 ± 0.055
Server aggregation	0.042 ± 0.004	0.189 ± 0.043
Global evaluation	2.144 ± 0.073	4.547 ± 0.227
Update preparation	0.045 ± 0.004	0.127 ± 0.015
Detector fitting	13.629 ± 0.467	45.029 ± 1.671
Detector scoring	0.278 ± 0.011	0.856 ± 0.035
Flag decision	0.000 ± 0.000	0.027 ± 0.024
All detector work	13.952 ± 0.476	46.040 ± 1.695
Combined compute	50.186 ± 1.943	127.239 ± 5.478

Our historical training, aggregation, detection and trap-generation components: not separately recorded. Network transmission: not measured in either local simulator. STD has no traps. All detector work includes preparation, fitting, scoring and decisions; it overlaps those table rows and must not be added a second time.

Table 2: Native-workflow elapsed compute, not a controlled speedup experiment. "Projected" rows are estimates.

Dataset	Workflow / horizon	Rounds	Mean s/round	Sum (min)
MNIST	Ours, measured warm-up	10	37.00	6.17
MNIST	STD, measured	10	53.22	8.87
MNIST	Ours, measured full horizon	100	34.41	57.35
MNIST	STD, measured full horizon	100	50.19	83.64
CIFAR-10	Ours, measured warm-up	10	117.76	19.63
CIFAR-10	STD, measured	10	125.23	20.87
CIFAR-10	Ours, measured full horizon	100	84.85	141.41
CIFAR-10	STD, measured full horizon	100	127.24	212.06

Table 3: Our 100 measured rounds grouped by events. Within-category mean is not detector-only overhead; amortized contribution is category sum / 100.

Dataset	Round category	Count	Within-category mean (s)	Contribution (s/round)
MNIST	Warm-up	10	37.00	3.70
MNIST	Probe without cycle decision	27	34.29	9.26
MNIST	Full-cycle decision	3	34.45	1.03
MNIST	Single audit	6	33.95	2.04
MNIST	Ordinary	54	34.04	18.38
CIFAR-10	Warm-up	10	117.76	11.78
CIFAR-10	Probe without cycle decision	63	82.04	51.68
CIFAR-10	Full-cycle decision	7	84.26	5.90
CIFAR-10	Single audit	2	76.76	1.54
CIFAR-10	Ordinary	18	77.51	13.95

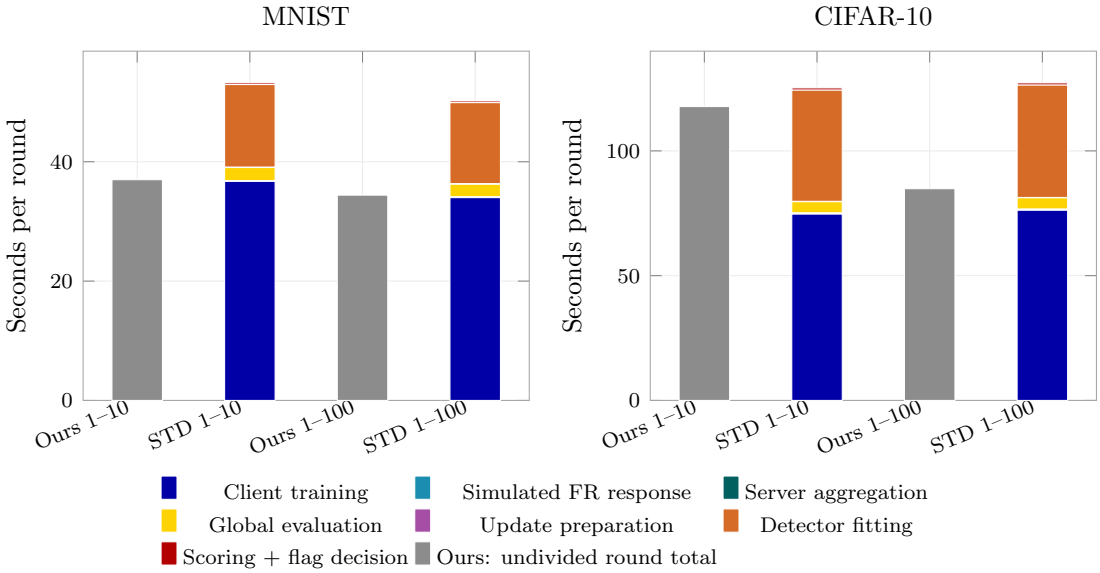


Figure 1: Timing breakdown: non-IID FR3, seed 42, 100 initial clients, 40% free riders. STD-DAGMM segments are measured; our gray bars are UNDIVIDED totals, not zero-cost detection. Ours 1–10 is warm-up only; ours 1–100 includes probing and removal. STD 1–100 is the mean of 100 actually executed rounds; first-ten-round bars use the first ten observations of these full runs, not the older pilot. Different trajectories, timer coverage and host load prevent a causal speedup claim.

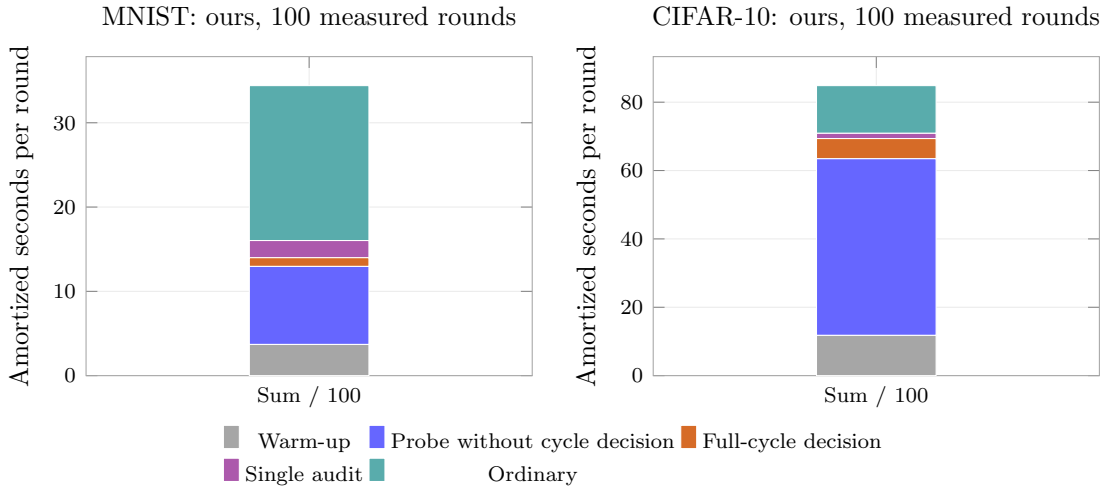


Figure 2: Amortizing our actual 100-round cost: each segment is the TOTAL time of one round category divided by 100. These are event categories, NOT isolated training/detection/trap CPU costs. Full-cycle rounds include training as well as decisions; collection/features can occur in other rounds.

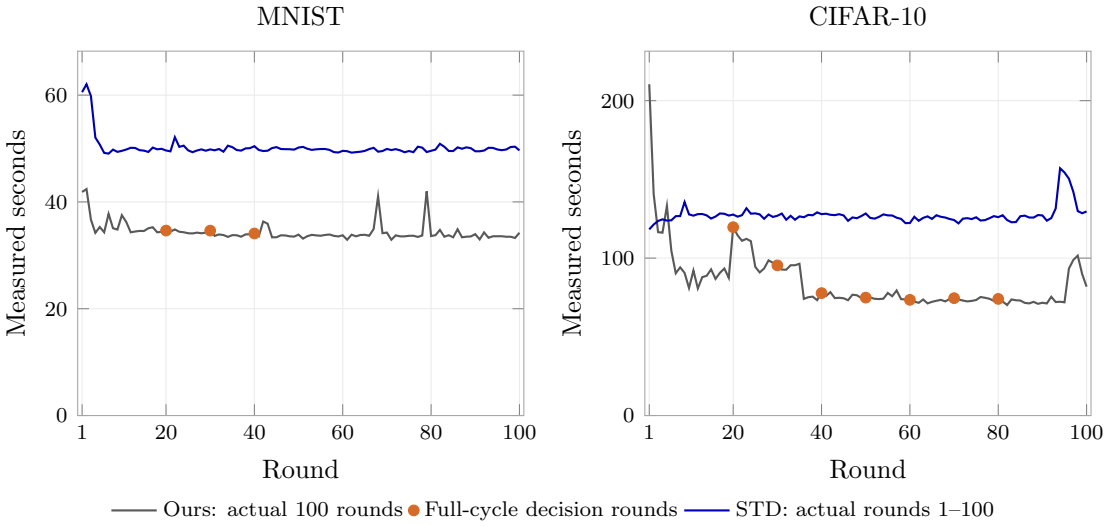


Figure 3: Actual per-round durations. Our 100-round trajectories include changing active-client counts and probing schedules. STD-DAGMM also has 100 measured rounds; neither trace is extrapolated. Markers indicate full-cycle decisions, not their isolated computational cost. Axis scales differ by dataset.